At any orientation, the angle of diagonal to the field is . In and , the currents are

The torque needed for uniform rotation is the net magnetic torque on the circuit, so

Substituting into gives

Group scores: : ; : ; : ; : total; : ; : ; : .

Method

The induced emfs in and are

The induced emfs in and are

With , shorted, only , act. If , are opened, the circuit is as shown. The currents in and are

The currents in and are

By symmetry, the potential differences across and across are